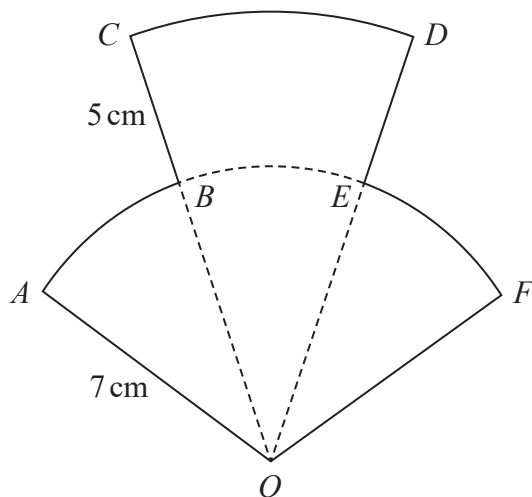


19 Here is a logo, $OABCDEF$.



$OABEF$ is a sector of a circle, centre O and radius 7 cm.

$OBCDE$ is a sector of a circle, centre O and radius 12 cm.

Angle $AOF = 100^\circ$

Angle $COD = 40^\circ$

The perimeter of the logo is P cm.

Find the exact value of P .

Give your answer in the form $a + b\pi$ where a and b are integers.

(Total for Question 19 is 4 marks)